

Excel Functions

A **function in Excel** is a **predefined formula** used to perform calculations, analyze data, or manipulate information efficiently.

It takes **inputs (arguments)**, processes them, and returns a result. Instead of writing long manual formulas, functions help us **solve problems quickly, accurately, and efficiently**.

Formula vs Function in Excel

Before learning more about functions, it is important to understand the difference between a **formula** and a **function**, as both are closely related.

What is a Formula?

A **formula** in Excel is an expression written by the user to perform calculations using:

- Cell references
- Numbers
- Operators (+, -, *, /)

Example

Instead of writing:

=A1 + A2 + A3 + A4 + A5 # Formula

We use:

=SUM(A1:A5) # Function

Excel automatically adds all values in the range.

Key Differences Between Formula and Function

Feature	Formula	Function
Definition	User-defined calculation	Predefined formula in Excel
Complexity	Can be long and manual	Simple and structured
Example	=A1 + A2 + A3	=SUM(A1:A3)
Usage	Basic calculations	Advanced and efficient operations

“Every function is a formula, but not every formula is a function.”

- Formula → You create it

- Function → Excel provides it

Why Do We Use Functions?

Functions help us to:

- Perform calculations faster
- Reduce manual effort
- Avoid errors
- Work with large datasets
- Make formulas easy to understand

Structure of a Function

Every function follows a standard format:

=FUNCTION_NAME(arguments)

Example:

=AVERAGE(A1:A10)

- **Function Name** → AVERAGE
- **Arguments** → A1:A10

Types of Functions

Based on their purpose, Excel functions can be grouped into the following categories

1. Data Analysis Functions

Purpose:

Used to **analyze, process, and transform** data

These functions help in calculations, summarization, and advanced data handling.

Sub-categories under Data Analysis:

1.1 Mathematical Functions - Used for numeric calculations

Functions:

- SUM() – Add values
- PRODUCT() – Multiply values
- ROUND(), ROUNDUP(), ROUNDDOWN() – Control decimals
- ABS() – Absolute value
- MOD() – Remainder
- POWER(), SQRT()
- CEILING(), FLOOR()

Use Case: Financial calculations, pricing, totals

1.2 Statistical Functions - Used to summarize and understand data

Functions:

- AVERAGE()
- MEDIAN()
- MODE()
- MAX() / MIN()
- COUNT(), COUNTA(), COUNTBLANK()
- STDEV()

Use Case: Reports, performance analysis

1.3 Array (Data Handling) Functions - Used to work with multiple values simultaneously

Functions:

- SUMPRODUCT()
- FILTER()
- UNIQUE()
- SORT()
- SORTBY()
- SEQUENCE()

Use Case: Dynamic reports, dashboards

2. Data Retrieval Functions

Purpose:

Used to **fetch, filter, and make decisions based on data**

These functions help in extracting specific information from datasets.

Sub-categories under Data Retrieval

2.1 Conditional Functions - Used to perform calculations based on conditions

Functions:

- SUMIF() / SUMIFS()
- COUNTIF() / COUNTIFS()
- AVERAGEIF() / AVERAGEIFS()

Use Case: Region-wise sales, filtered analysis

2.2 Logical Functions - Used for decision-making

Functions:

- IF()

- IFS()
- AND()
- OR()
- NOT()

Use Case: Pass/Fail, eligibility logic

2.3 Lookup & Reference Functions - Used to retrieve specific data from tables

Functions:

- VLOOKUP()
- HLOOKUP()
- XLOOKUP()
- INDEX()
- MATCH()
- OFFSET()

Use Case: Fetch employee details, product prices

3. Data Cleaning & Formatting Functions

Purpose:

Used to **clean, format, and standardize data**

Functions:

Text Functions:

- CONCAT() / TEXTJOIN()
- LEFT(), RIGHT(), MID()
- LEN()
- TRIM()
- UPPER(), LOWER(), PROPER()
- SUBSTITUTE()

Date & Time Functions:

- TODAY()
- NOW()
- DATE()
- YEAR(), MONTH(), DAY()
- DATEDIF()
- EOMONTH()

Use Case: Data preprocessing, reporting

4. Utility Functions

Purpose:

Used for supporting tasks like testing and simulation

Functions:

- RAND()
- RANDBETWEEN()

Use Case: Sample data generation

Mathematical Functions

Mathematical functions in Excel are used to perform numeric calculations and transformations.

They help in handling numbers efficiently for tasks like totals, rounding, pricing, and calculations.

SUM

Description: Adds a range of numbers.

Syntax: =SUM(number1, [number2], ...)

Example: =SUM(A1:A5)

PRODUCT

Description: Multiplies numbers together.

Syntax: =PRODUCT(number1, [number2], ...)

Example: =PRODUCT(A1:A3)

ROUND

Description: Rounds a number to a specified number of digits.

Syntax: =ROUND(number, num_digits)

Example: =ROUND(123.456,2)

ROUNDUP

Description: Rounds a number up, away from zero.

Syntax: =ROUNDUP(number, num_digits)

Example: =ROUNDUP(123.451,2)

ROUNDDOWN

Description: Rounds a number down, towards zero.

Syntax: =ROUNDDOWN(number, num_digits)

Example: =**ROUNDDOWN**(123.459,2)

ABS

Description: Returns the absolute (positive) value of a number.

Syntax: =ABS(number)

Example: =**ABS**(-10)

MOD

Description: Returns the remainder after division.

Syntax: =MOD(number, divisor)

Example: =**MOD**(10,3)

POWER

Description: Raises a number to a given power.

Syntax: =POWER(number, power)

Example: =**POWER**(2,3)

SQRT

Description: Returns the square root of a number.

Syntax: =SQRT(number)

Example: =**SQRT**(16)

CEILING

Description: Rounds a number up to the nearest multiple.

Syntax: =CEILING(number, significance)

Example: =**CEILING**(23,5)

FLOOR

Description: Rounds a number down to the nearest multiple.

Syntax: =FLOOR(number, significance)

Example: =**FLOOR**(23,5)

INT

Description: Rounds a number down to the nearest integer.

Syntax: =INT(number)

Example: =**INT**(5.9)

Statistical Functions

AVERAGE

Description: Calculates the average (mean) of a range of numbers.

Syntax: =AVERAGE(number1, [number2], ...)

Example: =AVERAGE(A1:A10)

MEDIAN

Description: Returns the middle value in a dataset.

Syntax: =MEDIAN(number1, [number2], ...)

Example: =MEDIAN(A1:A9)

MODE

Description: Returns the most frequently occurring value.

Syntax: =MODE(number1, [number2], ...)

Example: =MODE(A1:A10)

MAX

Description: Returns the largest value in a range.

Syntax: =MAX(number1, [number2], ...)

Example: =MAX(A1:A10)

MIN

Description: Returns the smallest value in a range.

Syntax: =MIN(number1, [number2], ...)

Example: =MIN(A1:A10)

COUNT

Description: Counts the number of cells containing numeric values.

Syntax: =COUNT(value1, [value2], ...)

Example: =COUNT(A1:A10)

COUNTA

Description: Counts all non-empty cells (text, numbers, etc.).

Syntax: =COUNTA(value1, [value2], ...)

Example: =COUNTA(A1:A10)

COUNTBLANK

Description: Counts empty cells in a range.

Syntax: =COUNTBLANK(range)

Example: =COUNTBLANK(A1:A10)

STDEV

Description: Estimates the standard deviation (data spread).

Syntax: =STDEV(number1, [number2], ...)

Example: =STDEV(A1:A10)

LARGE

Description: Returns the k-th largest value in a dataset.

Syntax: =LARGE(array, k)

Example: =LARGE(A1:A10, 2) # Returns 2nd largest value

SMALL

Description: Returns the k-th smallest value.

Syntax: =SMALL(array, k)

Example: =SMALL(A1:A10, 2) # Returns 2nd smallest value

Array (Data Handling) Functions

SUMPRODUCT

Description: Multiplies corresponding values in arrays and returns the sum of those products.

Syntax: =SUMPRODUCT(array1, [array2], ...)

Example: =SUMPRODUCT(A1:A3, B1:B3)

FILTER (Excel 365 / Excel 2021)

Description: Extracts data from a range based on a given condition.

Syntax: =FILTER(array, include, [if_empty])

Example: =FILTER(A1:B10, B1:B10>50)

UNIQUE (Excel 365 / Excel 2021)

Description: Returns unique values from a dataset (removes duplicates).

Syntax: =UNIQUE(array)

Example: =UNIQUE(A1:A10)

SORT (Excel 365 / Excel 2021)

Description: Sorts a range or array dynamically.

Syntax: =SORT(array, [sort_index], [sort_order])

Example: =SORT(A1:A10)

SORTBY (Excel 365 / Excel 2021)

Description: Sorts data based on another column or range.

Syntax: =SORTBY(array, by_array, [sort_order])

Example: =SORTBY(A1:B10, B1:B10, -1)

SEQUENCE (Excel 365 / Excel 2021)

Description: Generates a sequence of numbers automatically.

Syntax: =SEQUENCE(rows, [columns], [start], [step])

Example: =SEQUENCE(5)

TRANSPOSE (Excel 365 / Excel 2021)

Description: Converts rows into columns and columns into rows.

Syntax: =TRANSPOSE(array)

Example: =TRANSPOSE(A1:A3)

If A1:A3 is vertical, TRANSPOSE converts it into horizontal.

Text Functions

CONCAT

Description: Combines multiple text values into one.

Syntax: =CONCAT(text1, [text2], ...)

Example: =CONCAT(A1," ",B1)

TEXTJOIN

Description: Combines text using a delimiter.

Syntax: =TEXTJOIN(delimiter, ignore_empty, text1, ...)

Example: =TEXTJOIN(" ", TRUE, A1:A3)

LEFT

Description: Extracts characters from the left side.

Syntax: =LEFT(text, num_chars)

Example: =LEFT(A1,3)

RIGHT

Description: Extracts characters from the right side.

Syntax: =RIGHT(text, num_chars)

Example: =RIGHT(A1,2)

MID

Description: Extracts characters from the middle.

Syntax: =MID(text, start_num, num_chars)

Example: =MID(A1,2,3)

LEN

Description: Counts total number of characters.

Syntax: =LEN(text)

Example: =LEN(A1)

TRIM

Description: Removes extra spaces from text.

Syntax: =TRIM(text)

Example: =TRIM(A1)

UPPER

Description: Converts text to uppercase.

Syntax: =UPPER(text)

Example: =UPPER(A1)

LOWER

Description: Converts text to lowercase.

Syntax: =LOWER(text)

Example: =LOWER(A1)

PROPER

Description: Capitalizes first letter of each word.

Syntax: =PROPER(text)

Example: =PROPER(A1)

SUBSTITUTE

Description: Replaces specific text with new text.

Syntax: =SUBSTITUTE(text, old_text, new_text)

Example: =SUBSTITUTE(A1,"old","new")

Lookup & Reference Functions (Data Retrieval – Additional)

XLOOKUP (Modern Excel)

Description:

Searches for a value in a range and returns the corresponding result from another range. It is a more powerful and flexible replacement for VLOOKUP and HLOOKUP.

Syntax:

=XLOOKUP(lookup_value, lookup_array, return_array, [if_not_found], [match_mode], [search_mode])

Example:

=XLOOKUP(A2, A1:A10, B1:B10)

Looks for value in A2 within A1:A10 and returns the corresponding value from B1:B10.

OFFSET

Description:

Returns a reference to a cell or range by moving a specified number of rows and columns from a starting point.

Syntax: =OFFSET(reference, rows, cols, [height], [width])

Example: =OFFSET(A1, 2, 1)

Starts from A1, moves 2 rows down and 1 column right → returns value from B3.

Utility Functions

RAND

Description: Generates a random decimal number between 0 and 1.

Syntax: =RAND()

Example: =RAND()

Returns a value like 0.5487 (changes every time the sheet recalculates)

RANDBETWEEN

Description: Generates a random integer between two specified numbers.

Syntax: =RANDBETWEEN(bottom, top)

Example: =RANDBETWEEN(1,100)

Returns a random number between 1 and 100

